Elasticity/Distensibility—The ability to return to the original resting length after a muscle has been stretched.

9.8 Skeletal system and its function



The skeletal system is constituted by a framework of bones and cartilages. It is derived from the embryonic **mesoderm**. Muscles are attached to the bones by means of tendons and provide the necessary force required for the bones of the skeleton to operate as levers. There are three types of skeletal systems. They are,

Hydrostatic skeleton, which is found in soft-bodied invertebrates. It is a fluid filled-cavity encircled by muscles (e.g. Earth worm).

Exoskeleton, which is found in invertebrates. It is a rigid hard case present outside the body of animals (e.g. Cockroach).

Endoskeleton, which is found inside the body of vertebrates. It is composed of bones and cartilages, surrounded by muscles. (eg. Human being).

In human beings, the skeletal system is made up of 206 bones and cartilages. It is grouped into two principal divisions – the **axial skeleton** and the **appendicular skeleton**. The axial skeleton consists of 80 bones and the appendicular skeleton consists of 126 bones (Figure 9.6 and Table.1).

Functions of skeletal system

- Support –It forms a rigid framework and supports the weight of the body against gravity.
- Shape - It provides and maintains the shape of the body.

All muscles produce movement, but only skeletal muscle is responsible for locomotion. What is meant by this statement?

- Protection – It protects the delicate internal organs of the body.
- Acts as reservoir – It stores minerals such as calcium and phosphate. Fat (Triglyceride) is stored in yellow bone marrow and represents a source of stored energy for the body.
- Locomotion – It acts as lever along with the muscles attached to it.
- Strength – It can withstand heavy weight and absorbs mechanical shock.
- As a haemopoietic tissue – Red and White blood cells are produced in the bone marrow of the ribs, spongy bones of vertebrae and extremities of long bones.

9.9 The Axial skeleton

Axial skeleton forms the main axis of the body. It consists of the skull, hyoid bone, vertebral column and thoracic cage.

a) The Skull

The skull is composed of two sets of bones – cranial and facial bones. It consists of 22 bones of which 8 are cranial bones and 14 are facial bones (Figure 9.5). The cranial bones form the hard protective outer covering of the brain and called the brain box. The capacity of the cranium is 1500 cm^{3}. These bones are joined by sutures which are immovable. They are a **paired parietal, paired temporal** and **individual bones** such as the **frontal, sphenoid, occipital and ethmoid**.

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